

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$L_{izm.} = (0,51 \pm 0,05) \text{ H}$$

$$R = (270 \pm 10) \Omega$$

$$C_1 = 0,5 \mu\text{F}$$

$$C_2 = 1,0 \mu\text{F}$$

$$C_3 = 2,0 \mu\text{F}$$

$$\left. \begin{array}{l} C_1 = 0,5 \mu\text{F} \\ C_2 = 1,0 \mu\text{F} \\ C_3 = 2,0 \mu\text{F} \end{array} \right\} \pm 0,1 \mu\text{F} ?$$

$$\omega_{0,1} = \sqrt{\frac{1}{LC_1}}$$

 $\Rightarrow$

$$\boxed{\begin{array}{l} L_1 = \frac{1}{\omega_{0,1}^2 C_1} \\ L_2 = \frac{1}{\omega_{0,2}^2 C_2} \\ L_3 = \frac{1}{\omega_{0,3}^2 C_3} \end{array}}$$

~~$$\omega_{0,1} = 2268 \frac{1}{\text{s}}$$~~
~~$$\omega_{0,2} =$$~~
~~$$\omega_{0,3} =$$~~

~~$$\omega_{0,1} = 2,27 \pm 0,11$$~~

$$\omega_{0,1} = (2300 \pm 100) \frac{\text{rad}}{\text{s}}$$

$$\omega_{0,2} = (1570 \pm 50) \frac{\text{rad}}{\text{s}}$$

$$\omega_{0,3} = (1100 \pm 80) \frac{\text{rad}}{\text{s}}$$

iz fita

$$U(\omega) = \frac{U_0}{\sqrt{1 + \left(\frac{\omega_0 L}{R}\right)^2 \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)^2}}$$

fit ~ grafih

$$\frac{\omega_0 L}{R} = Q$$

$$\Rightarrow L_1 = \frac{1}{(2300 \frac{1}{\text{s}})^2 \cdot 0,5 \mu\text{F}} = 0,378 \text{ H}$$

$$\Rightarrow L_{1\text{err}} = \frac{1}{(4\%)^2 \cdot 20\%} = \frac{1}{28\%} = 28\%$$

$$\Rightarrow L_{2\text{err}} = (3\%)^2 \cdot 10\% = 16\%$$

$$\Rightarrow L_{3\text{err}} = (7\%)^2 \cdot 5\% = 19\%$$

$$\left[\begin{array}{l} L_1 = (0,4 \pm 0,1) \text{ H} = 0,378 (1 \pm 0,28) \text{ H} \\ L_2 = (0,41 \pm 0,06) \text{ H} = 0,406 (1 \pm 0,16) \text{ H} \\ L_3 = (0,41 \pm 0,08) \text{ H} = 0,413 (1 \pm 0,19) \text{ H} \end{array} \right] \quad \begin{array}{l} \bar{L} = 0,399 (1 \pm 0,13) \\ \bar{L} = (0,40 \pm 0,09) \text{ H} \end{array}$$